Unit 12: The Laplace Transform

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following twelve problems in order. They climb through the defining integral, linearity and the table, the first shift, the derivative rule on a full IVP, the integral/t-multiplication rules, inverse transforms by partial fractions and by completing the square, a repeated root, a complete five-step solution, Heaviside shifting forward and back, and a Dirac-delta impulse-response capstone. Solutions follow in Part III.

1. **(From the definition)** Using only the defining integral $\int_0^\infty e^{-st} f\,dt$, compute $\mathcal{L}\{e^{3t}\}$ and state the half-plane of convergence.
2. **(Linearity and the table)** Find $\mathcal{L}\{3t^2 - 2\cos 4t + 5\}$.
3. **(First shift theorem)** Find $\mathcal{L}\{e^{-2t}\sin 3t\}$.
4. **(Derivative rule on an IVP)** Transform $y'' + 4y = 0$ with $y(0) = 1$, $y'(0) = 0$, solve for $Y(s)$, and invert to find $y(t)$.
5. **(t-multiplication rule)** Find $\mathcal{L}\{t\,e^{2t}\}$ using $\mathcal{L}\{tf(t)\} = -F'(s)$.
6. **(Inverse — distinct roots)** Find $\mathcal{L}^{-1}\left\{\dfrac{s+7}{(s-1)(s+3)}\right\}$.
7. **(Inverse — complete the square)** Find $\mathcal{L}^{-1}\left\{\dfrac{1}{s^2+4s+13}\right\}$.
8. **(Inverse — repeated root)** Find $\mathcal{L}^{-1}\left\{\dfrac{3s+1}{(s+2)^2}\right\}$.
9. **(Full five-step IVP)** Solve $y'' + 3y' + 2y = 0$ with $y(0) = 1$, $y'(0) = 0$ by the Laplace method.
10. **(Heaviside shifting, forward)** Find $\mathcal{L}\{u(t-3)\cos(t-3)\}$, then find $\mathcal{L}\{u(t-2)\,t^2\}$ by first re-expressing t^2 in $(t-2)$.
11. **(Heaviside shifting, inverse)** Find $\mathcal{L}^{-1}\left\{\dfrac{e^{-2s}}{s^2}\right\}$.
12. **(Dirac impulse response, capstone)** Solve $y'' + 4y = \delta(t-\pi)$ with $y(0) = 0$, $y'(0) = 0$, and describe the motion before and after the kick.

Part III — Complete Solutions

Solution 1. Merge the two exponentials and integrate:

$$\int_0^\infty e^{-st}e^{3t}\,dt = \int_0^\infty e^{-(s-3)t}\,dt = \left[-\tfrac{1}{s-3}e^{-(s-3)t}\right]_0^\infty = \frac{1}{s-3}.$$

The boundary term at ∞ vanishes only when $s-3>0$, so the transform converges for $s>3$.

Solution 2. Apply linearity term by term with $\mathcal{L}\{t^2\}=2/s^3$, $\mathcal{L}\{\cos 4t\}=s/(s^2+16)$, $\mathcal{L}\{1\}=1/s$:

$$\mathcal{L}\{3t^2-2\cos 4t+5\} = \frac{6}{s^3} - \frac{2s}{s^2+16} + \frac{5}{s}.$$

Solution 3. Start from $\mathcal{L}\{\sin 3t\} = \dfrac{3}{s^2+9}$ and apply the first shift $s \mapsto s+2$:

$$\mathcal{L}\{e^{-2t}\sin 3t\} = \left.\frac{3}{s^2+9}\right|_{s\to s+2} = \frac{3}{(s+2)^2+9}.$$

The completed-square denominator $(s+2)^2+9$ is the fingerprint of a damped oscillation.

Solution 4. Transform with $\mathcal{L}\{y''\}=s^2Y-sy(0)-y'(0)$ and substitute $y(0)=1$, $y'(0)=0$:

$$(s^2Y-s)+4Y=0 \;\Rightarrow\; (s^2+4)Y=s \;\Rightarrow\; Y(s)=\frac{s}{s^2+4}.$$

This is exactly $\mathcal{L}\{\cos 2t\}$, so

$$y(t)=\cos 2t.$$

Solution 5. With $F(s)=\mathcal{L}\{e^{2t}\}=\dfrac{1}{s-2}$, apply $\mathcal{L}\{tf\}=-F'(s)$:

$$\mathcal{L}\{te^{2t}\} = -\frac{d}{ds}\left(\frac{1}{s-2}\right) = -\left(-\frac{1}{(s-2)^2}\right) = \frac{1}{(s-2)^2}.$$

Equivalently, this is the first shift applied to $\mathcal{L}\{t\}=1/s^2$.

Solution 6. Decompose $\dfrac{s+7}{(s-1)(s+3)} = \dfrac{A}{s-1}+\dfrac{B}{s+3}$. Clearing denominators, $s+7=A(s+3)+B(s-1)$. Set $s=1$: $8=4A \Rightarrow A=2$. Set $s=-3$: $4=-4B \Rightarrow B=-1$:

$$\frac{s+7}{(s-1)(s+3)} = \frac{2}{s-1} - \frac{1}{s+3} \implies \mathcal{L}^{-1} = 2e^t - e^{-3t}.$$

Solution 7. Complete the square in the denominator: $s^2+4s+13=(s+2)^2+9$. This matches the shifted sin with $a=-2$, $\omega=3$, and a $1/\omega$ scale:

$$\frac{1}{(s+2)^2+3^2} = \frac{1}{3}\cdot\frac{3}{(s+2)^2+3^2} \implies \mathcal{L}^{-1} = \frac{1}{3}\,e^{-2t}\sin 3t.$$

Solution 8. A repeated factor needs two terms, $\dfrac{3s+1}{(s+2)^2} = \dfrac{A}{s+2}+\dfrac{B}{(s+2)^2}$. Then $3s+1=A(s+2)+B$, so $A=3$ and $B=1-2A=-5$:

$$\frac{3}{s+2} - \frac{5}{(s+2)^2} \implies \mathcal{L}^{-1} = 3e^{-2t} - 5t\,e^{-2t} = (3-5t)\,e^{-2t}.$$

The te^{-2t} term is the signature of the repeated real root.

Solution 9. Transform and substitute $y(0) = 1$, $y'(0) = 0$:

$$(s^2Y - s) + 3(sY - 1) + 2Y = 0 \;\Rightarrow\; (s^2 + 3s + 2)Y = s + 3.$$

Factor $s^2 + 3s + 2 = (s + 1)(s + 2)$ and decompose $\dfrac{s+3}{(s+1)(s+2)} = \dfrac{A}{s+1} + \dfrac{B}{s+2}$ with $A = 2$, $B = -1$:

$$Y(s) = \frac{2}{s+1} - \frac{1}{s+2} \implies y(t) = 2e^{-t} - e^{-2t}.$$

Solution 10. The first part is the second shift applied to $\mathcal{L}\{\cos t\} = \dfrac{s}{s^2+1}$ with $c = 3$:

$$\mathcal{L}\{u(t-3)\cos(t-3)\} = e^{-3s}\,\frac{s}{s^2+1}.$$

For the second part, the argument t^2 must be rewritten in $(t-2)$ before the theorem applies: $t^2 = (t-2)^2 + 4(t-2) + 4$. Transform each shifted piece ($\mathcal{L}\{t^2\} = 2/s^3$, $\mathcal{L}\{t\} = 1/s^2$, $\mathcal{L}\{1\} = 1/s$) and attach e^{-2s}:

$$\mathcal{L}\{u(t-2)\,t^2\} = e^{-2s}\left(\frac{2}{s^3} + \frac{4}{s^2} + \frac{4}{s}\right).$$

Solution 11. The factor e^{-2s} signals a delay by $c = 2$. Since $\mathcal{L}^{-1}\{1/s^2\} = t$, the second shift gives

$$\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2}\right\} = u(t-2)\,(t-2).$$

The response is a ramp that stays flat at zero until it switches on at $t = 2$.

Solution 12. Transform with $\mathcal{L}\{\delta(t-\pi)\} = e^{-\pi s}$ and zero initial data:

$$(s^2 + 4)Y = e^{-\pi s} \;\Rightarrow\; Y(s) = e^{-\pi s}\,\frac{1}{s^2+4}.$$

Since $\mathcal{L}^{-1}\{1/(s^2+4)\} = \frac{1}{2}\sin 2t$, the second shift gives

$$y(t) = \frac{1}{2}\,u(t-\pi)\,\sin\big(2(t-\pi)\big) = \frac{1}{2}\,u(t-\pi)\,\sin 2t.$$

The mass sits at rest until the impulse at $t = \pi$, which jolts the velocity and launches free oscillation thereafter — the impulse response switched on at the kick.

Unit Key Takeaway

Every problem above is the same loop: transform, do the algebra, invert. The derivative rule carries the initial data into the equation, completing the square handles complex roots, e^{-cs} tracks every delay, and $\mathcal{L}\{\delta\} = e^{-cs}$ turns an impulse into a clean algebraic term.